Structure-Dependent Few-Shot Segmentation of Abdominal MRI: A Reptile Meta-Learning Approach

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Abstract. Accurately segmenting anatomical structures from abdominal magnetic resonance (MR) images is critical for clinical applications such as disease assessment and treatment planning. When only a small number of annotated examples are available for a new structure, conventional supervised methods fail to generalise due to their reliance on large, labelled datasets. This work investigates few-shot medical image segmentation using Reptile, a first-order meta-learning algorithm, compared against a U-Net trained from scratch under identical low-data conditions. Both models are evaluated across 1-shot, 3-shot, and 5-shot settings on two unseen anatomical structures. Reptile consistently outperformed the baseline across all conditions, achieving a mean DSC of 0.369 on task_1 compared to 0.259 for the baseline, confirming that meta-learned initialisation provides a meaningful advantage in low-data regimes. Performance on task_8 was near-zero for both methods, highlighting that structural properties such as size, shape variability, and foreground sparsity are primary determinants of few-shot segmentation difficulty. The effect of shot count was modest and non-monotonic, suggesting that support set size and adaptation budget must be considered jointly.

Keywords: Few-shot learning · Medical image segmentation · Meta-learning · Abdominal MRI · Reptile

1 Introduction

Segmentation of anatomical structures in abdominal magnetic resonance imaging (MRI) is a critical component of modern clinical workflows, supporting applications such as diagnosis, treatment planning, and disease monitoring. Accurate organ delineation is particularly important in cases involving complex or rare pathologies, where precision directly affects clinical outcomes. However, manual segmentation is labour-intensive, time-consuming, and subject to inter-observer variability, placing a significant burden on radiologists and limiting scalability in clinical practice [6].

Recent advances in deep learning have significantly improved automated medical image segmentation. Architectures like U-Net have become the de facto standard due to their ability to capture both global context and fine structural details [8, 4, 2]. Despite their success, these models rely heavily on large annotated datasets. Obtaining such annotations requires expert clinicians and substantial time. This issue is further exacerbated for rare anatomical structures or novel clinical targets with limited labelled examples.

Few-shot learning has emerged as a promising approach to address this data scarcity by enabling models to generalise from a small number of labelled samples [10]. Rather than learning each task independently, few-shot methods leverage prior knowledge to rapidly adapt to new tasks with minimal data. A key framework for enabling few-shot learning is meta-learning, often described as "learning to learn." Meta-learning approaches train models across a distribution of tasks such that they acquire an initialisation that can be efficiently adapted to unseen tasks using only a few gradient updates [3].

While Model-Agnostic Meta-Learning (MAML) demonstrates strong performance, it requires complex second-order gradient computations. In contrast, Reptile provides a simpler first-order alternative, learning effective initialisations for rapid adaptation without second-order derivatives [7]. In this work, we adopt a Reptile-based meta-learning approach for few-shot segmentation of abdominal MRI data, using U-Net as the backbone architecture due to its proven effectiveness in medical image segmentation.

2 Methods

Dataset and tasks. An abdominal MRI dataset provided by the module organisers is used, comprising 8 anonymised structures each treated as a separate segmentation task. For each task, 40 scans are converted to 2D greyscale axial slices and resized to 256×256 pixels. Tasks are split at the task level into 4 train, 2 validation, and 2 test tasks to avoid structure leakage. For a task and shot count n, support slices are sampled without replacement using seed 42 and all remaining slices form the query set, giving fully reproducible episodes.

Model, Loss and Metric. A compact U-Net is employed for binary segmentation. The encoder uses four blocks (32, 64, 128, 256 channels) with 2×2 max pooling, mirrored by the decoder with transposed convolutions and skip connections. Group Normalisation (8 groups) replaces Batch Normalisation [11] for stability with small batches. Training optimises a combined loss $L = 0.2L_{BCE} + 0.8L_{Dice}$, which handles foreground sparsity better than cross-entropy alone. Performance is evaluated using the Dice Similarity Coefficient (DSC) on query slices with a 0.2 binarization threshold.

Meta-training and adaptation. A Reptile first-order meta-learning algorithm is used to learn an initialisation that can be rapidly adapted to new tasks. At each outer iteration, a training task is sampled uniformly, a clone of the meta-model is adapted via the inner loop, and the Reptile update $\theta \leftarrow \theta + \epsilon_t(\phi - \theta)$ is applied, where ϕ are the adapted weights. SGD is used in the inner loop to avoid task-specific optimiser state that would distort this finite-difference update [7].

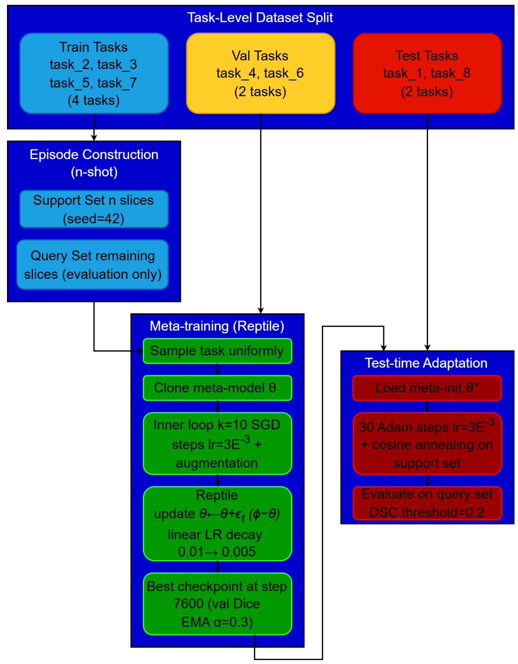

Fig. 1. Overview of the Reptile meta-training and few-shot adaptation pipeline. Tasks are split at the task level; episodes are constructed by sampling n support slices.

For meta-training, we use SGD (learning rate 3×10^{-3} with linear decay to 0.005) for $k = 10$ inner steps and 8,000 outer steps, applying random flips and $\pm 15°$ rotations. The best checkpoint (step 7,600) is selected via an EMA ($\alpha = 0.3$) smoothed validation Dice every 200 steps. At test time, the meta-initialisation is adapted on the support set of each test task using 30 Adam steps (learning rate 1×10^{-3}, cosine annealing) without augmentation. To reduce sensitivity to randomness, the full pipeline is evaluated across three seeds (42, 7, 123).

3 Experiments

Setup and Evaluation Protocol. Experiments were conducted on two held-out test tasks from the abdominal MRI dataset: task_1 and task_8. Each task was evaluated under three few-shot regimes: 1-shot, 3-shot, and 5-shot, corresponding to the number of labelled support slices available at adaptation time. The Reptile meta-learner was compared against a *baseline*, defined as a U-Net trained from scratch on the same n support slices, using the same random seeds and the same number of inner adaptation steps (30), but without any meta-initialisation. This isolates the contribution of the Reptile meta-learning procedure from the effect of the adaptation itself.

Table 1. Mean DSC ± standard deviation (3 seeds) for Reptile vs. baseline.

Task	Method	1-shot	3-shot	5-shot
1	Reptile	0.360 ± 0.009	0.399 ± 0.025	0.349 ± 0.005
1	Baseline	0.249 ± 0.036	0.286 ± 0.034	0.242 ± 0.057
8	Reptile	0.051 ± 0.036	0.049 ± 0.030	0.063 ± 0.043
8	Baseline	0.013 ± 0.009	0.017 ± 0.013	0.032 ± 0.019

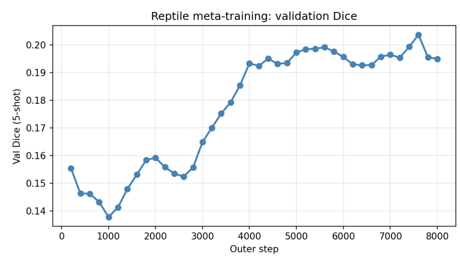

Fig. 2. Reptile meta-training validation Dice (5-shot, 10 adaptation steps) evaluated on held-out validation tasks every 200 outer steps.

Compute & Checkpointing. For all test evaluations, the best Reptile checkpoint, selected as the outer training step yielding the highest validation Dice on held-out validation tasks (step 7,600, val Dice 0.203), was used rather than the final checkpoint, consistent with early-stopping practice. All experiments were run on a single NVIDIA A40 GPU via a Jupyter notebook; the codebase was additionally validated to run on a Google Colab T4 GPU.

4 Results

Reptile consistently outperforms the baseline across all conditions. On task_1, the meta-initialisation yields a mean advantage of +0.113 DSC. Task_8 is markedly harder for both methods (Reptile mean DSC 0.054), likely due to a smaller, less contiguous target structure with greater inter-patient shape variability.

The meta-training validation curve (Figure 2) shows rapid improvement in val Dice to step ∼3,000, followed by a plateau around 0.19–0.20, with the best checkpoint selected at step 7,600. Note that val Dice is computed using only 10 adaptation steps on the held-out validation tasks and serves solely as a checkpoint selection criterion; it is not directly comparable to the test DSC in Table 1, which uses 30 adaptation steps on unseen test tasks.

Qualitative results (Figure 3) reflect the quantitative findings. On task_1, Reptile produces a spatially coherent activation localised around the correct region; the baseline is noisier with greater spread into background tissue. On task_8, both methods produce diffuse, near-zero activations, confirming that 5

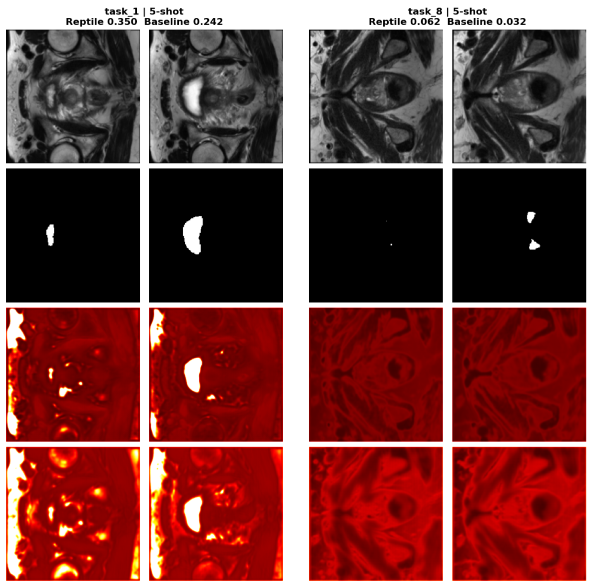

Fig. 3. Qualitative segmentation results at 5-shot for the two held-out test tasks. Each grouped panel shows two representative query slices. Rows from top: MR image, ground truth mask, Reptile prediction, baseline prediction.

support examples are insufficient to identify the target anatomy regardless of initialisation quality.

5 Discussion

Reptile vs. Baseline. The results confirm that meta-learned initialisation provides a consistent and meaningful advantage over training from scratch in few-shot abdominal MRI segmentation. Across all shot counts and both test tasks, Reptile outperforms the baseline, with the gap most pronounced on task_1. This is consistent with the Reptile objective of finding an initialisation in a region of parameter space amenable to rapid adaptation [7].

Task Difficulty and Structure Variability. The performance difference between task_1 and task_8 (mean DSC 0.369 vs. 0.054) demonstrates that anatomical structure properties substantially modulate few-shot segmentation difficulty. Task_1 achieves reasonable Dice scores across all conditions, likely corresponding to a larger structure with relatively consistent cross-sectional appearance across patients, providing sufficient signal for adaptation from a small support set. Task_8, by contrast, produces near-zero Dice for both methods across all shot counts, consistent with a smaller or less contiguous structure exhibiting greater inter-patient shape and contrast variability. These findings suggest that structure size and shape consistency are stronger determinants of few-shot segmentation success than the choice of initialisation strategy, and that a minimum level of structural regularity may be required for meta-learning to be effective.

Effect of Shot Count. Increasing support slices does not produce consistent improvements within the 30-step adaptation budget. On task_1, performance peaks at 3-shot (DSC 0.399) and declines at 5-shot (0.349), while task_8 remains flat. Additional slices likely increase the adaptation loss landscape's complexity without a proportional increase in convergence steps. This suggests gains from additional labelled examples may only materialise if the adaptation step count scales accordingly.

Limitations. Evaluation is restricted to two test tasks, limiting generalisability, particularly for task_8 where near-zero DSC scores make variance estimates unreliable. The non-monotonic shot count effect warrants investigation across wider adaptation budgets. Additionally, the fixed binarization threshold of 0.2 may disadvantage task_8 where predicted probabilities are low.

6 Conclusion

Reptile-based meta-learning was applied to few-shot segmentation of abdominal MRI structures, and shown to consistently outperform a from-scratch baseline across all evaluated conditions, with a mean DSC advantage of +0.113 on task_1 [7]. These results confirm that meta-learned initialisations provide a practical benefit in low-data medical imaging settings, even under constrained adaptation budgets.

The primary limitation identified is the failure of both methods on task_8, which is attributed to the combination of a small, sparse target structure, high inter-patient shape variability, and severe foreground-background imbalance. This represents a broader challenge in few-shot medical image segmentation: meta-learning can accelerate adaptation but cannot substitute for an informative support set or a well-calibrated loss function. The non-monotonic effect of shot count further highlights that support set size and adaptation budget must be considered jointly rather than in isolation.

Future work should address these failure modes. Adopting sparse-structure losses, such as focal [5] or Tversky loss [9], would better balance foreground-background gradients. Augmenting the training distribution with MRI-specific transformations would improve robustness to inter-patient variability [8]. Finally, meta-regularisation techniques [1] could improve meta-training stability, yielding initialisations that transfer more reliably to structurally atypical tasks.

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